PAIDIPILLI PURUSHOTHAM

ECE Student: Full-Stack Software Engineering, .NET, & Angular, React , IoT Data Pipelines

23pa1a04b4@vishnu.edu.in | +91-8093853830

Bhimavaram, AP | LinkedIn | GitHub | YouTube

EDUCATION

B.Tech in Electronics and Communication Engineering (ECE) — Aug 2023 – Aug 2027
Vishnu Institute of Technology (CGPA: 9.73/10 till 6th sem) — *Bhimavaram, AP*

Intermediate (MPC) – Aditya Junior College (Marks: 971/1000) — *Jun 2021 – Apr 2023*

TECHNICAL SKILLS

- **Programming Languages:** C Programming, Basics of Java, C#, Basics of C++, Python, HTML/CSS
- **Hardware & HDLs:** VLSI, Verilog, RISC-V Architecture, Embedded C
- **Tools & EDA:** Synopsys VCS, Dc-Shell, Pt-Shell, Custom Compiler, TCAD, Cadence Virtuoso, OpenSource (RTL-to-GDSII), Xilinx Zynq / Vivado, LabVIEW
- **Frameworks & Technologies:** .NET, Angular, IoT, SignalR, WebRTC, FPGA
- **Development & Workflow:** Git (CLI / Command Line Interface), GitHub

PROJECTS

Arogya Connect | Live | Code — *Angular 17+, ASP.NET Core, SignalR, PostgreSQL*

- Developed a real-time hospital discovery platform supporting 5 distinct user roles (Patient, Doctor, Nurse, Staff, Admin) with strict role-based access control.
- Implemented a SignalR-based smart emergency alert system routing SOS alerts to correct network hospitals in under 30 seconds, solving the 60% patient information gap.
- Incorporated Firebase OAuth and custom JWT authentication alongside a robust MediatR and Repository pattern backend architecture to secure data transmission.

Sure Proed LMS & Video Platform | Live | Code — *React, Django, Redis, WebRTC*

- Engineered a comprehensive Learning Management System (LMS) integrating secure video conferencing capabilities to support 100+ concurrent users with sub-50ms latency.
- Built distinct, role-based dashboards tailored for Admins, Mentors, and Students to efficiently manage, schedule, and join live classes and sessions.

Context-Aware Hazard Monitor | Code — *Angular, Firebase, Chart.js, C++*

- Architected a full-stack dashboard utilizing Firebase Authentication and enforcing strict one-to-one device-to-user binding via Firebase Security Rules.
- Implemented a real-time telemetry pipeline to visualize multi-sensor data streams (MQ-2, DHT11, LDR) with sub-200 ms synchronization latency.
- Programmed a local rule-based virtual assistant that evaluates live telemetry against 4 safety thresholds, providing immediate operator guidance with <10ms response time.

EXPERIENCE

Full-Stack .NET Developer Trainee — Jun 2024 – Aug 2024
Software Training — *Bhimavaram, AP*

- Completed an intensive .NET full-stack program, mastering C#, ASP.NET Core Web API, and Entity Framework Core for robust backend architecture.
- Applied clean architecture and MediatR design patterns, culminating in the end-to-end development of the Arogya Connect smart hospital platform.

Embedded Systems & IoT Intern — Oct 2025 – Mar 2026
Sure Proed (Technical Training Trust) — *Puttaparthi, AP*

- Programmed Context-Aware Text-to-Speech algorithms in Embedded C, achieving 98% translation accuracy and minimizing latency for real-time edge processing.
- Engineered telemetry pipelines interfacing C++ endpoints with cloud databases to visualize multi-sensor data streams.

Summer Research Intern — May 2026 – Jul 2026
IIT Bhubaneswar (Dept. of ECE) — *Bhubaneswar, Odisha*

- Selected from 1,000+ applicants for an exclusive offline summer internship under Dr. Himanshu Padole to conduct advanced computational research.

VLSI Engineering Trainee — Aug 2024 – Present
HCL — *Remote*

- Formulated the SafeALU architecture, optimizing computational logic to achieve a 25% decrease in energy usage during processing operations.

KEY ACHIEVEMENTS

- **1st Runner-Up (Spark Tank):** Secured an INR 30k cash prize among 50+ teams from E-Cell VITB for the AQMD project.
- **1st Place (Hustlethon 2k25):** Engineered the winning IoT Aquaculture Water Testing solution out of 100+ competitors.
- **Advanced RISC-V Tapeout:** Completed an intensive 10-week RTL-to-GDSII flow for a 32-bit RISC-V core, ranking top 22 nationally. *[View GitHub Repo]*
- **HAM Radio Operator:** Hold a General License; actively operate at the Vishnu Space Applications Center.
- **Certifications:** HCL VLSI Foundation | NPTEL Elite Comm. Systems | VLSI SoC Design | NPTEL Elite+ IoT